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INHIBITORY FUNCTION IN THE STIMULUS-RESPONSE COMPATIBILITY TASK [1,2]

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Summary.—Changes of a location-based inhibitory function were investigated by performing a Stimulus-Response Compatibility task under two conditions. This task allows study of the efficiency of the inhibitory function by analyzing differences in error rates and in response time under various conditions of stimulus and response compatibility. In Exp. 1, with 28 college students, the effect of a dual task on the Stimulus-Response Compatibility effect was examined. In a dual-task, a predetermined stimulus such as a color is identified and the number of times a color is displayed is counted. In this experiment, under a dual task using visual stimuli, a decline in error responses and reduction of the Stimulus-Response Compatibility effect were observed. In Exp. 2, 29 college students participated in an identical experiment with the exception that an auditory stimulus was presented as the dual task. Similar to Exp. 1, there was a significant reduction in the number of error responses in the dual-task condition. Conversely, there was no significant change in Stimulus-Response Compatibility effect as estimated by response time. These results suggest that the two visual pathways as proposed by Goodale (1995) and the evocation of intentional attention may affect the Stimulus-Response Compatibility effect.

This study investigated changes in the inhibitory function (Luria, 1961) using a Stimulus-Response Compatibility task (Fitts & Seeger, 1953; Hommel & Prinz, 1997). In such a task, a stimulus is displayed on a screen either to the left or to the right of a fixation point. Participants are instructed to press the left response-button when a stimulus is displayed on the right and to press the right response-button when a stimulus is displayed on the left of the fixation point. Typically in this task, longer response times are observed for contralateral stimulus and response-button combinations in comparison to ipsilateral combinations. This exemplifies the adaptability of location relations between a stimulus and a response (Kornblum, Hasbroucq, & Osman, 1990).

The Stimulus-Response Compatibility task is also known as the directional Stroop task (Diamond, 2002), and it has been studied as a case of response inhibition (Christ, White, Mandernach, & Keys, 2001). This is because in order to emit a contralateral button-press response, the ipsilateral response for the same stimulus must be inhibited. Similar to letter names in

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the Stroop task (Stroop, 1935), the Stimulus-Response Compatibility task forces participants to inhibit an easy response to make the desired one. Differences in response times between ipsilateral stimulus response pairs and contralateral ones are known as the Stimulus-Response Compatibility effect, which is considered to be an index of the efficiency of the inhibitory function. Short response times indicate an efficient inhibitory function, whereas long ones indicate an inefficient function. Studies of age-related changes in the Stimulus-Response Compatibility effect have shown that response times are longer for elderly people, indicative of an age-related decrease in inhibitory function (Christ, *et al.*, 2001). In addition, it is known that the Stimulus-Response Compatibility effect is lower in 15-yr.-old children than in adults, suggesting that the inhibitory function might be developing in children of this age.

In general, it has been shown that the Stimulus-Response Compatibility effect becomes larger, indicative of a decrease in the efficiency of inhibitory function, when the task is conducted under a dual-task condition. For example, the Stimulus-Response Compatibility effect and error rates increased under a dual-task condition in studies using an antisaccade task, which is one type of SRC task (Roberts, Hager, & Heron, 1994). It is interesting that a decrease of processing resources is closely related to inhibitory function. This finding also corroborates evidence that the Stimulus-Response Compatibility effect increases in elderly people, who may have decreased processing capacity.

The present study was originally planned to investigate the enhancement of the Stimulus-Response Compatibility effect under the dual-task condition. Quite unexpectedly, however, a decrease in the effect was observed for young participants under a dual-task condition.

Experiment 1

Recently, several aspects of the inhibitory function have been investigated. For example, the inhibitory function has been separated into location-based and identity-based subtypes (Connelly & Hasher, 1993; May, Kane, & Hasher, 1995). The antisaccade task (Butler, Zacks, & Henderson, 1999) is an example of location-based inhibitory function. In this task, the eyes are forced to move to the opposite side by inhibiting eye movement to the right side to match a stimulus presented on the right. This task requires inhibition of the response to match a place where a stimulus is indicated. The Stroop task is an example of identity-based inhibitory function. In this task, a color is named by ignoring prepotent letter information, such as when the letters of the word 'red' are printed in ink of a different color. According to this system of classification, the inhibitory function in the Stimulus-Response Compatibility task falls within the location-based category.

Current theorizing on the processing of visual stimuli assumes two visual pathways (Ungerleider & Mishkin, 1982; Goodale, 1995). The ventral pathway is assumed to process the characteristic information regarding the stimulus, the so-called 'what' stream, and the dorsal pathway, assumed to process stimulus location information, the so called 'where' stream. It is considered that to a certain extent these two types of processing are conducted independently in the two pathways.

In this study, a dual-task was used to identify color as the stimulus characteristic to examine the decrease in inhibitory function when this dual-task adds an extra processing load on a concurrent stimulus-location Stimulus-Response Compatibility task. According to the discussion above, it is assumed that the extra load incurred by the task concerning stimulus-location information would be less severe if the two visual pathways are independent.

Method

Participants.—Right-handed college students ($n=28$; 8 men and 20 women; M age 22.5 yr., $SD=3.9$) with normal vision participated.

Equipment.—Equipment used in the experiment consisted of a DOS/V PC (PC9821nb10) for controlling the experiment, a CRT display (Akia RT145WX) for displaying the stimuli, and round response buttons (Jonan Electric Industrial).

Procedure.—In a Compatible condition, participants responded to the location identification task individually by pressing a selected response-button ipsilateral to an indicated stimulus. First, a fixation point was shown at the center of the visual field. Then, a red or a green circle with a diameter of 4.5 mm was shown randomly either on the left or the right of the fixation point at a visual angle of 10.7°. The distance from the fixation point to the participant was approximately 40 cm. The probability of the stimulus appearing on either side was set at 50%. Participants were instructed to keep their hands on both response-buttons throughout the experiment and to press the ipsilateral response-button accurately and as soon as possible when a stimulus was detected; then the trial was terminated. The next trial was initiated following a randomly selected Reaction-Stimulus Interval of either 500 msec., 1500 msec., or 2500 msec. Three different Reaction-Stimulus Intervals were used to avoid anticipation by the participants. After eight practice trials, two experimental blocks (one block = 16 trials) were presented with a 5-sec. interval between the blocks. In the Incompatible condition, the contralateral response-button to the indicated stimulus was pressed. Number of trials and Reaction-Stimulus Interval in the Incompatible condition were identical to those of the Compatible condition.

Both Compatible and Incompatible conditions were practiced under single-task and dual-task conditions. The dual-task condition consisted of reporting the number of 'red' stimuli among all displayed stimuli after completing each block. 'Red' stimuli were set to appear seven, eight, or nine times in one block of 16 trials. Under the dual-task condition, participants pressed the selected response-button depending on the location of the indicated stimulus, and this task was loaded with the additional task of counting and remembering the number of times stimuli of a certain color appeared.

Therefore, this experiment consisted of 2 within subject factors (Compatible and Incompatible conditions) × task (single-task and dual-task) design. Each participant completed a total of 128 trials, 32 trials in each condition. The conditions were counterbalanced between participants. Response latency and the number of error responses were recorded as the dependent variables. The mean response latency was calculated for each participant in each condition.

Results

Response latency and error ratios are shown on Table 1. It is clear from these results that the Stimulus-Response Compatibility effect both in the single task and the dual task were confirmed in this experiment because mean response time in the Incompatible condition was longer than that in the Compatible condition. Under the single-task condition, the mean response time in the Compatible condition was 410 msec. ($SD = 31$), and in the Incompatible condition it was 492 msec. ($SD = 67$), a difference of 83 msec. Conversely, on the whole, the response latency under the dual-task condition was longer than under the single task; however, it was 504 msec. ($SD = 97$) in the Compatible condition[3] and 555 msec. ($SD = 127$) in the Incompatible condition, showing a smaller difference of 51 msec. The analysis of variance indicated a significant main effect between the single-task and the dual-task conditions ($F_{1,27} = 18.78$, $p < .01$) and between the Compatible and Incompatible conditions ($F_{1,27} = 49.33$, $p < .01$).

For further confirmation, the difference in response latency between Compatible and Incompatible conditions was examined using the procedure described by Hartley and Kieley (1995), in which the difference between mean response latencies in the Compatible and Incompatible conditions were calculated and divided by the individual response latency in the Compatible condition. This analysis further confirmed that the Stimulus-Response Compatible effect decreased significantly under the dual-task condition compared to the single-task condition ($F_{1,27} = 10.57$, $p < .01$).

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[3]In the study by of Roberts, *et al.* (1995) using an antisaccade task, no response delay was observed for the Compatible condition even under the dual-task condition, whereas there was a response delay for the Incompatible condition. The difference in results of the Roberts, *et al.* and current studies may be related to the different responses used in these. The manual task in this study requires more exercise than the eye-movement task so the dual task may have been effective even under the Compatible condition.

TABLE 1

MEANS AND STANDARD DEVIATIONS FOR REACTION TIMES (MSEC.) AND ERROR RATES (%) IN TWO EXPERIMENTS

Exp.	Task	Condition	Reaction Time		Error Rate	
			M	*SD*	*M*	*SD*
1	Single	Compatible	410[a]	31	1.1[b]	2.9
		Incompatible	492	67	2.5	3.5
	Dual	Compatible	504	97	0.4	1.1
		Incompatible	555	128	1.1	2.5
2	Single	Compatible	407[c]	31	1.1[d]	2.9
		Incompatible	489	67	2.4	3.0
	Dual	Compatible	489	107	0.1	0.6
		Incompatible	562	137	1.2	2.3

[a]Significant effects for compatibility ($p<.01$), concurrent task ($p<.01$), and compatibility × concurrent task ($p<.05$). [b]Significant effects for compatibility ($.05<p<.10$) and concurrent task ($p<.05$). [c]Significant effects for compatibility ($p<.01$) and concurrent task ($p<.01$). [d]Significant effects for compatibility ($p<.05$) and concurrent task ($p<.05$).

The error analysis in the single-task condition indicated a mean error response ratio of 1.1% (*SD*=2.9) in the Compatible condition and 2.5% (*SD*=3.5) in the Incompatible condition. In the dual-task condition it was 0.4% (*SD*=0.4) in the Compatible condition and 1.1% (*SD*=2.5) in the Incompatible condition. Although there were only a few error responses, on the whole, their number decreased significantly in the dual task.

Discussion

It has been suggested that the inhibitory function was more effective under a dual-task condition given the Stimulus-Response Compatibility effect or the difference between Compatible and Incompatible conditions. Error rate also decreased under the dual-task condition indicating that the inhibitory function was strong in this condition. What is the reason for this tendency in spite of the increased load in the dual-task condition? It may be possible that the dual task of identifying a stimulus characteristic was independent of the Stimulus-Response Compatibility task even at the level of visual pathways. The task load may not have increased in the dual-task condition because the two tasks were processed independently of each other. Furthermore, it is possible that the impulsive response to a location, i.e., pressing the ipsilateral response-button in response to an ipsilateral stimulus in the Incompatible condition, was suppressed by the additional load of generating intentional attention to a stimulus.

Moreover, in Exp. 1, over all mean response latency increased whereas mean error rate decreased, indicative of a speed-accuracy trade off (Henmon, 1911). Finger-pointing-and-call is a method used in Japanese industry to prevent errors. In this method the operator points to a button or to a target to confirm and then vocalizes about the situation, before pressing the button or operating the target. This is an example of the speed-accuracy trade off. By pointing to a target to direct intentional attention, impulsive responses are restrained and error responses are reduced, even though the response time increases (Haga, Akatsuka, & Shiroto, 1996). The stimulus color identification task used in Exp. 1 might have produced the same effect as finger-pointing-and-call.

Experiment 2

Exp. 2 was conducted to confirm the results of Exp. 1 by changing the content of the dual task. In Exp. 2, the procedure was identical to Exp. 1, with the exception that the dual task consisted of an auditory stimulus. While performing the Stimulus-Response Compatibility task, an auditory stimulus was presented a number of times. After each block, the participants' reported the number of times the auditory stimulus was heard. An auditory task was chosen to measure changes in the Stimulus-Response Compatibility effect when stimuli of a different modality from the Stimulus-Response Compatibility task are used in the dual task. It was predicted that the Stimulus-Response Compatibility effect would increase in Exp. 2 as a result of using auditory stimuli because intentional attention to visual stimuli would decrease in the Stimulus-Response Compatibility task.

Method

Participants.—Right-handed college students ($n=29$; 8 men and 21 women; M age 21.5 yr., $SD=3.5$) with normal vision participated in the study.

Equipment.—Equipment used were the same as those of Exp. 1.

Procedure.—The procedure was identical to that of Exp. 1, with the exception that an auditory stimulus (90 db) was randomly presented as the dual-task stimulus, 7 to 9 times during each 16-trial block for 200 msec., similar to the visual stimuli in Exp. 1. The sound was emitted from a speaker placed under the center of the monitor. The dual task in Exp. 2 was to report the number of times the auditory stimulus was presented at completion of each block.

Results

Response latencies and error rates are shown in Table 1. It is clear from these results that the Stimulus-Response Compatibility effect was confirmed in both the single-task and the dual-task condition because the response latency in the Incompatible condition was longer than in the Compatible condition. In the single-task condition, it was 407 msec. ($SD=33$) in the Compatible condition and 489 msec. ($SD=67$) in the Incompatible condition, a difference of 82 msec. Conversely, in the dual-task condition, it was 489 msec. ($SD=106$) in the Compatible condition and 563 msec. ($SD=135$) in the Incompatible condition, a difference of 74 msec. An analysis of variance indicated a significant main effect between the single-task and dual-task conditions ($F_{1,28} = 19.93$, $p < .01$) and between the Compatible and Incompatible conditions ($F_{1,28} = 33.38$, $p < .01$). However, there were no significant interactions ($F_{1,28} = 0.29$, ns), suggesting that there was no significant difference in the Stimulus-Response Compatibility effect between the single and dual tasks. For further confirmation, the differences in response latencies between the Compatible and the Incompatible conditions in the single and dual tasks were analyzed based on work of Hartley and Kieley (1995). This analysis indicated no significant difference in the Stimulus-Response Compatibility effect between Compatible and Incompatible conditions ($F_{1,28} = 1.85$, ns).

Mean error rates were 1.1% ($SD = 0.6$) in the Compatible condition and 2.4% ($SD = 3.0$) in the Incompatible condition in the single-task condition, whereas they were 0.1% ($SD = 0.6$) in the Compatible condition and 1.2% ($SD = 2.3$) in the Incompatible condition during the dual task. Although there were few error responses over all, error responses decreased during the dual task ($F_{1,28} = 5.20$, $p < .05$).

Discussion

There was no significant change in the Stimulus-Response Compatibility effect in mean response times between the single-task and the dual-task conditions; however, there was evidence of inhibitory function because there was a significant decrease in error rates in the dual-task condition. Similar to Exp. 1, it is possible that the inhibitory function increased by evoking intentional attention through stimulus identification.

It is possible that in the dual task one role of the identification task is to inhibit the impulsive response to an ipsilateral location; however, there were no significant differences in the Stimulus-Response Compatibility as estimated by response latencies. Therefore, in comparison to Exp. 1, in Exp. 2 the inhibitory function decreased given a shift of attention to auditory stimuli that were of a different modality from the stimuli in the Stimulus-Response Compatibility task.

General Discussion

The results of Exps. 1 and 2 suggest that it is possible to increase the location-based inhibitory function by using a stimulus-identification dual-task. In addition, results also suggest that the inhibitory function is more effective when the stimuli used in the dual task are in the same modality as in the Stimulus-Response Compatibility task. These results may point to the relation between an action pathway and a cognitive pathway in the visual system as was suggested by Goodale (1995). An action pathway is a pathway for 'action', originating from the primary visual cortex and extending through the posterior parietal cortex. A cognition pathway is a pathway for recognition of a stimulus, originating from the primary visual cortex and extending through the inferotemporal cortex. Activation of the cognition pathway might mediate the restraint of impulsive responses of an action pathway that acts without recognition. There is the possibility that error resulting from unconscious stimulus-matching responses was restrained by drawing attention to stimulus characteristics, even though these characteristics were not related to the current response, i.e., pressing a button. (1) In Exp. 1, the loading effect of a stimulus-identification task on the Stimulus-Response Compatibility task reinforced the inhibitory function. (2) In Exp. 2, changing the stimuli in the identification task to the auditory modality reduced the effect of inhibitory function. Based on these findings, it is possible to infer that a cognition pathway might have influenced the function of the action pathway, particularly with regard to reaction control.

Conversely, analysis of error rates suggested that the inhibitory function was reinforced even when the target of the identification task was auditory. This suggests the possibility that the inhibitory function of an action pathway might be reinforced through imposing a load to evoke intentional attention, even when the target of the Stimulus-Response Compatibility task is in a different modality. It is suggested that studies on the Stimulus-Response Compatibility effect should focus on the effect of stimuli in different modalities.

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